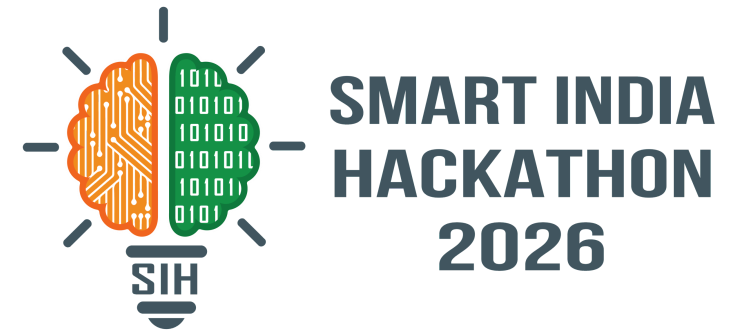


SMART INDIA HACKATHON 2026

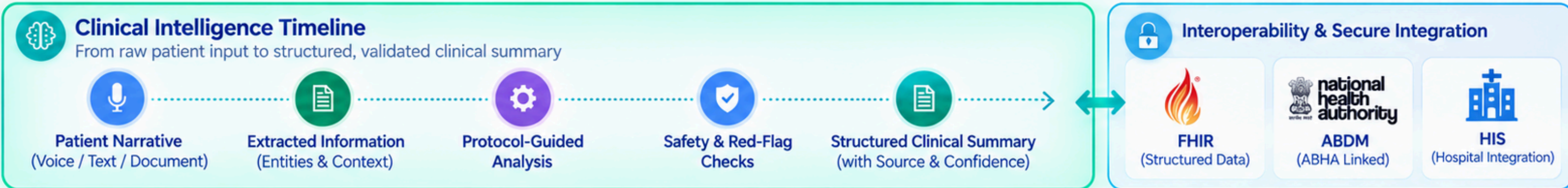


- **Problem Statement ID – SIH26047**
- **Problem Statement Title - Patient case taking software**
- **Theme - MedTech / BioTech / HealthTech**
- **PS Category - Software**
- **Team ID -**
- **Team Name - Codeholics**



Proposed Solution

From Patient's Story to Physician-Ready Summary



TECHNICAL APPROACH

From raw patient input to structured, actionable clinical data



Implementation Process

Input Collection



- Voice (Multi-language)
- Handwritten / printed documents
- Kiosk / Web interface

Preprocessing



- Speech-to-Text (ASR)
- OCR for documents
- Text cleaning & normalization

Clinical Processing



- Protocol-based question flow (SOCRATES, ROS, etc.)
- Medical entity extraction
- Red-flag detection

Validation & Structuring



- Confidence scoring
- Human-in-the-loop verification
- Structured clinical data generation

Output & Integration



- Structured patient summary
- Evidence linking (voice/docs)
- Export / Integrate with HIS / ABDM (FHIR R4)



Technologies Used



Frontend

React / Next.js
Responsive Web App



Backend

Python + FastAPI
Modular APIs



AI / ML

ASR (Indic languages)
OCR
NLP (Medical NER)
LLM (Summarization)



Database

PostgreSQL
Object Storage
(Audio / Documents)



Integration

FHIR R4
ABDM / HIS

FEASIBILITY AND VIABILITY

Practical. Scalable. Real-World Ready.



Feasibility Analysis

Built with existing technologies and domain knowledge.



Use of Proven Technologies

Speech-to-Text, OCR, NLP, LLMs (readily available and mature)



Built on Open Resources

Open datasets, medical ontologies (SNOMED CT, ICD-10), AYUSH resources



Modular and Scalable Architecture

Independent modules for easy scaling and deployment



Deployable in Real-World Settings

Can be used at PHCs, district hospitals and tertiary care centers



Pilot Ready

Feasible to implement as a prototype with minimal infrastructure



Potential Challenges and Risks

Identifying risks to build a stronger solution.



Variability in Handwriting

Different doctor styles, abbreviations and languages



Data Privacy and Security

Sensitive patient data needs strong protection



Integration with Existing Systems

Compatibility with hospital workflows (HIS/ABDM)



User Adoption

Resistance to change, training required for doctors and staff



Edge Cases and Ambiguity

Incomplete or unclear information in prescriptions and patient input



Strategies for Overcoming Challenges

Turning challenges into opportunities.



Train on Diverse Indian Datasets

Use real-world handwritten prescriptions, multi-language speech data



Privacy-by-Design

End-to-end encryption, role-based access, compliance with NDHM/ABDM guidelines



Standards-Based Integration

Use FHIR APIs and ABDM-compliant modules for seamless integration



Training and Change Management

Simple, intuitive UI and training support for healthcare staff



Continuous Feedback and Iteration

Human-in-the-loop validation and regular model improvements



Viable, Scalable and Sustainable

Leverages existing infrastructure, supports government healthcare initiatives (ABDM, HIS), and can be scaled across India for long-term impact.



From
Pilot to Pan-India

IMPACT AND BENEFITS

Better histories. Faster care. Healthier India.



Impact on Target Audience

- Reduces waiting time for doctors
- More accurate and complete patient histories
- Bridges communication gaps (multilingual)
- Improves patient experience and trust



Empowered Patients.
Efficient Doctors.



Benefits for Healthcare System

- Increases doctor efficiency and consultation quality
- Standardized and structured data
- Supports early detection of red flags
- Better integration with ABDM and HIS
- Helps in continuity of care across visits



Streamlined Workflows.
Stronger Healthcare.



Social & Economic Benefits

- Improved access to quality healthcare
- Reduced diagnostic errors
- Lower repeat visits and overall costs
- Supports rural and underserved populations
- Creates opportunities for local language AI solutions



Inclusive Care.
Greater Equity.



Environmental Benefits

- Reduces paper usage (digital records)
- Minimizes physical storage of documents
- Supports sustainable and green healthcare
- Lower resource consumption in hospitals



Greener Hospitals.
A Sustainable Tomorrow.

RESEARCH AND REFERENCES

Built on proven research, medical standards and real-world needs.



Clinical Guidelines

- NICE Guidelines
- WHO Clinical History Taking Principles
- ICMR Guidelines
- Standard History-Taking Frameworks (SOCRATES, ROS, etc.)



AYUSH References

- Ministry of AYUSH Guidelines
- CCRAS / CCRH Manuals
- Prakriti & Vikriti Assessment Standards
- Evidence-based AYUSH Research Papers



Technology & AI

- Speech Recognition (Indic Languages)
- OCR for Handwritten Text
- NLP & Medical NER
- Explainable AI
- LLM Safety & Guardrails
- Research Papers (ACL, EMNLP, etc.)



Links / References

- NHP – National Health Portal
- Ayush Research Portal
- ABDM Documentation
- FHIR Specifications
- PubMed Research Papers
- WHO Resources

